BECS 22451

Experiment No: 10

MATLAB SIMULINK FOR CONTROL

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Experiment No : 10

Experiment Name : MATLAB Simulink for Control

APPARATUS

- A PC or Laptop with MATLAB Simulink installed.

THEORY AND DIAGRAMS

Control System Simulation with Simulink:

Control systems play a pivotal role in regulating and influencing the behaviour of dynamic systems. Simulating these systems provides a valuable means to study and analyse their responses under different conditions.

Control Systems and Simulink:

Control systems are crucial in engineering to manage and manipulate the behavior of dynamic systems. Simulink, a graphical programming environment in MATLAB, facilitates the modeling, simulation, and analysis of these systems. It offers an intuitive platform for designing and visualizing control system models.

Second-Order System Model:

A second-order system is characterised by two poles in its transfer function, representing its dynamic response. In this experiment, we will construct a second-order system model using Simulink. The model's parameters, such as damping ratio and natural frequency, significantly influence the system's behaviour.

Step Input and System Response:

A step input is a fundamental test signal used to analyse the dynamic response of a system. When applied to a control system, it induces a sudden change, allowing us to observe how the system reacts over time. We can visualise and analyse the system's response to a step input through Simulink, gaining insights into its stability and transient characteristics.

Practical Scenario

Solar tracking is a technology designed to enhance the efficiency of solar energy capture by aligning solar panels or mirrors with the sun's position throughout the day. Unlike fixed solar installations, which remain in a static position, solar tracking systems dynamically adjust the orientation of solar components to maximise exposure to sunlight, as shown in Figure 1.

THEORY AND DIAGRAMS

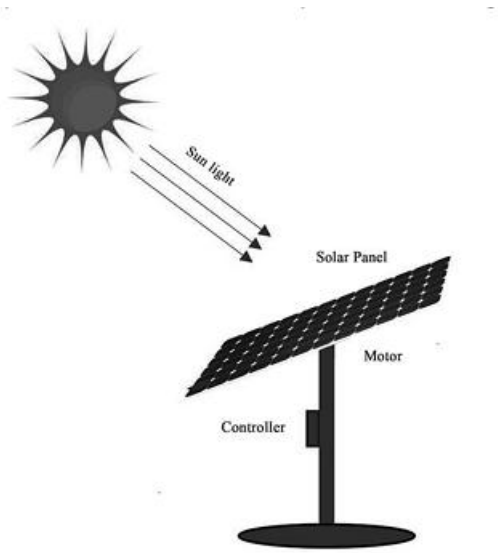


Figure 1

Motion of the Solar Panel:

The rotational motion of the solar panel can be described using the second-order differential equation (1) below.

$$\frac{d^2\theta}{dt^2} = \frac{1}{J}(T - K_d\frac{d\theta}{dt}) \qquad (1)$$

Where,

•T=$KgKti$

•$d2\theta/\ dt2$: Second derivative of the tracking angle with respect to time, representing the rotational dynamics of the solar panel.

•J: Moment of inertia of the solar panel. It quantifies the distribution of mass around the axis of rotation.

•Kd: Damping constant. It represents the damping or resistive forces acting against the rotation of the solar panel.

•T: Torque of the motor

•Kt : Torque constant, relating the motor current to the torque produced.

THEORY AND DIAGRAMS

Electrical Dynamics of the Motor:

The electrical dynamics of the motor can be described using the first-order linear differential equation (2) below.

$$\frac{di}{dt} = \frac{1}{L}\left(V - K_g K_f \frac{d\theta}{dt} - Ri\right) \qquad (2)$$

● di/dt: Rate of change of current with respect to time, representing the electrical dynamics of the motor.

● L: Inductance of the motor circuit. It measures the ability of the coil to induce a voltage in itself when the current through it changes.

● V: Applied voltage to the motor.

● Kg: Gain constant. It represents the motor gain, which relates the electrical input to the mechanical output.

● Kf : Feedback constant. It accounts for any feedback mechanism influencing the motor dynamics.

● $d\theta/dt$: Rate of change of the tracking angle with respect to time. It's a coupling term indicating that the motor's current is influenced by the rate of change of the tracking angle.

● R: Resistance in the motor circuit.

PID Control Systems:

PID (Proportional-Integral-Derivative) control optimises solar trackers by continuously adjusting panel orientation for maximum sunlight exposure. The proportional term (P) responds to current errors, integral (I) eliminates steady-state error over time, and derivative (D) anticipates future errors, preventing overshooting. This dynamic PID algorithm ensures precise tracking as the sun moves across the sky. It enhances tracking accuracy, adapts to environmental changes, and maximises energy harvesting. The result is an efficient and responsive solar tracker, crucial for optimising energy capture in solar power systems and contributing to increased overall energy efficiency.

THEORY AND DIAGRAMS

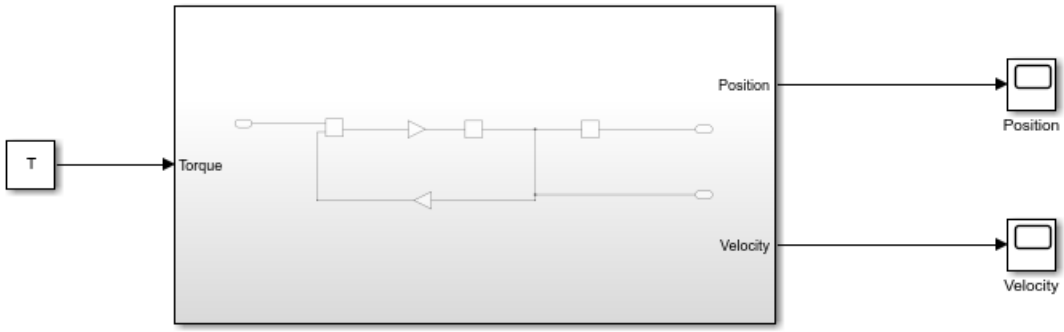


Figure 2

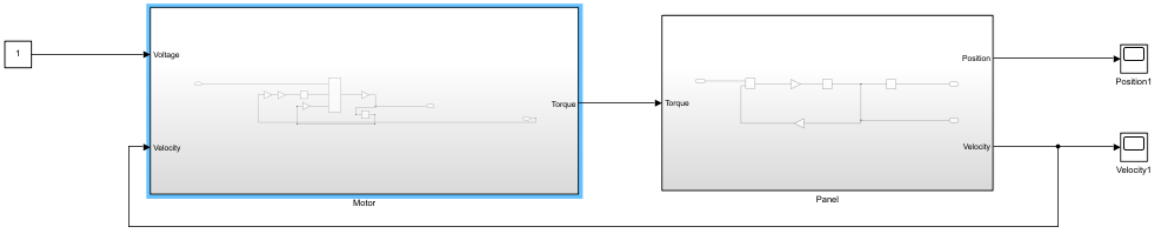


Figure 3

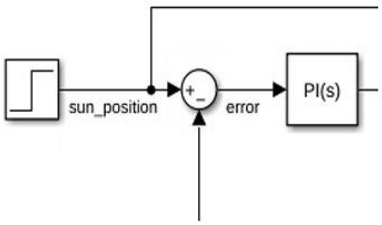


Figure 4

THEORY AND DIAGRAMS

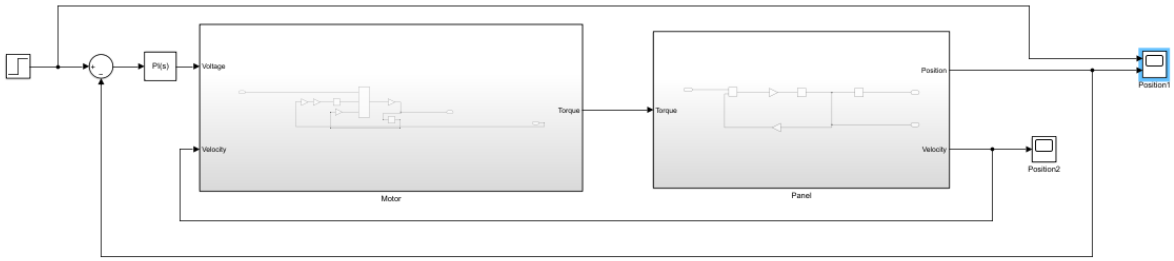


Figure 5

PROCEDURE

Part 01

1. MATLAB was started.

2. Simulink was opened, and a “Blank Module” was created in the simulation start window.

3. The Simulink project was saved as “part_01.slx” in a folder named “EC_2021_073.” A Word file was created and renamed with the student number, then saved in the same folder.

4. The Simulink model was designed to control the motion of the solar panel as shown in Figure 1, referring to equation (1). The following steps were followed:

a. A constant block from Sources was used as input, and its value was changed to 10, representing the torque input to the system.

b. An integrator block was used to integrate and obtain velocity and position from acceleration.

c. The damping constant Kd was defined in a gain block by clicking the three dots and creating Kd. Its value was set to 5 and the location to Base Workspace.

d. Similar steps as in 4(c) were followed for J, with a value of 8.6 and Base Workspace as the location.

e. A text annotation was added using the following method of writing to display the equation (1): **\frac{d^2\theta}{dt^2}=\frac{1}{J}(T-K_d\frac{d\theta}{dt}\right)**

f. Two scopes were attached to observe position and velocity. Components were selected, and a subsystem was created from the selection. The final model for the Motion of the Solar Panel resembled as shown in Figure 2.

g. Outputs were observed (using 10 for simulation stop time), and screenshots were taken.

PROCEDURE

Part 02

1. The Simulink model for the electrical dynamics of the motor was designed, referring to equation (2). The following instructions were followed:

a. Values were defined in gain blocks:

L: 1.0000e-05

Kg: 2000

Kf: 0.0700

R: 10

The location was set to Base Workspace.

b. A subsystem was created using the knowledge from Part 01, resulting in a model resembling Figure 3.

c. The model was frequently saved in the created folder.

2. A Step Function (Unit step) was inserted, a Sum block was added, and the sign list was changed to |+-. A PID controller block was added from the library browser and installed, as shown in Figure 4. A step function (Unit Step) was inserted, along with a Sum block with a |+- configuration.

3. The PID controller block was double-clicked, and values were changed. The controller was set to PI mode since the D term was unnecessary for slow sun movement.

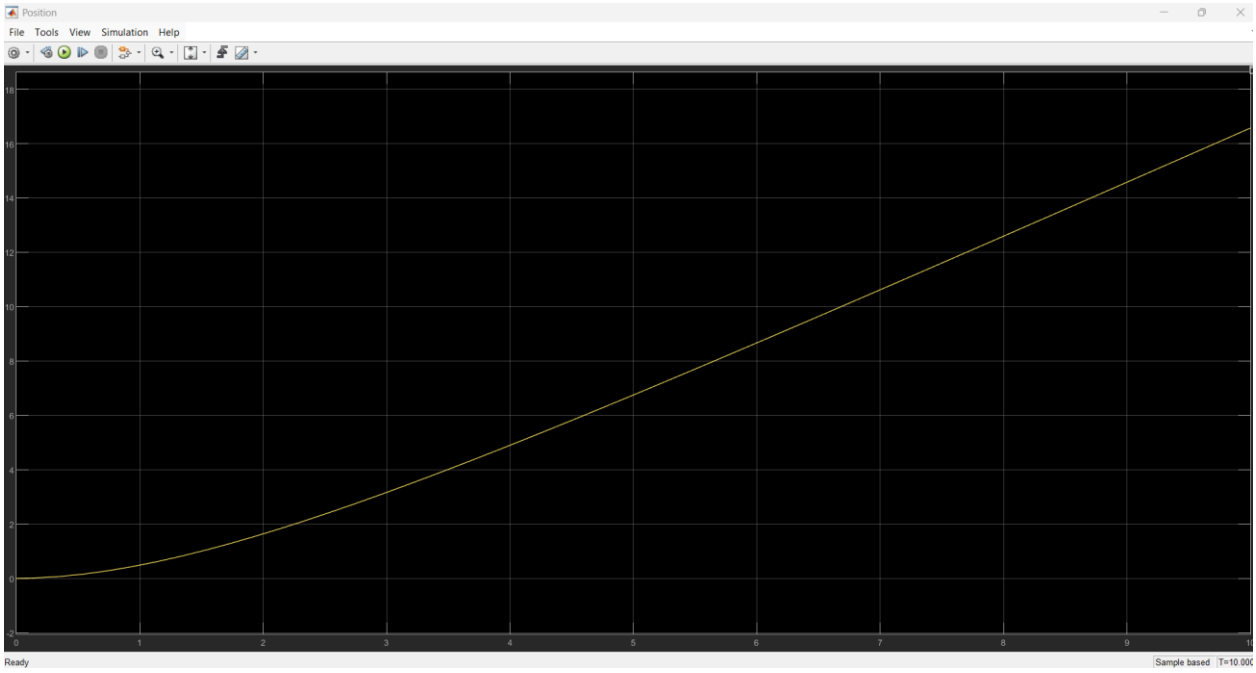
4. The final model was designed as shown in Figure 5. The position scope was used to observe the sun and solar panel positions

5. The simulation was run, and the outputs were observed using scopes.

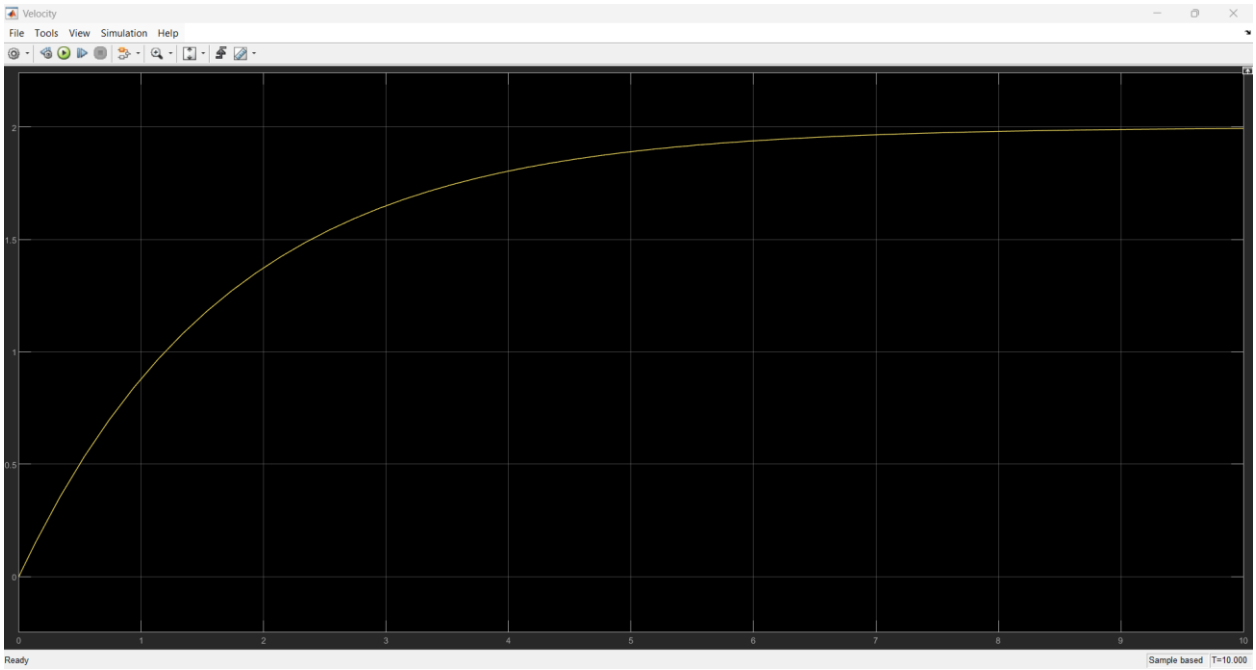
OBSERVATIONS

Part _01

Graph of Position



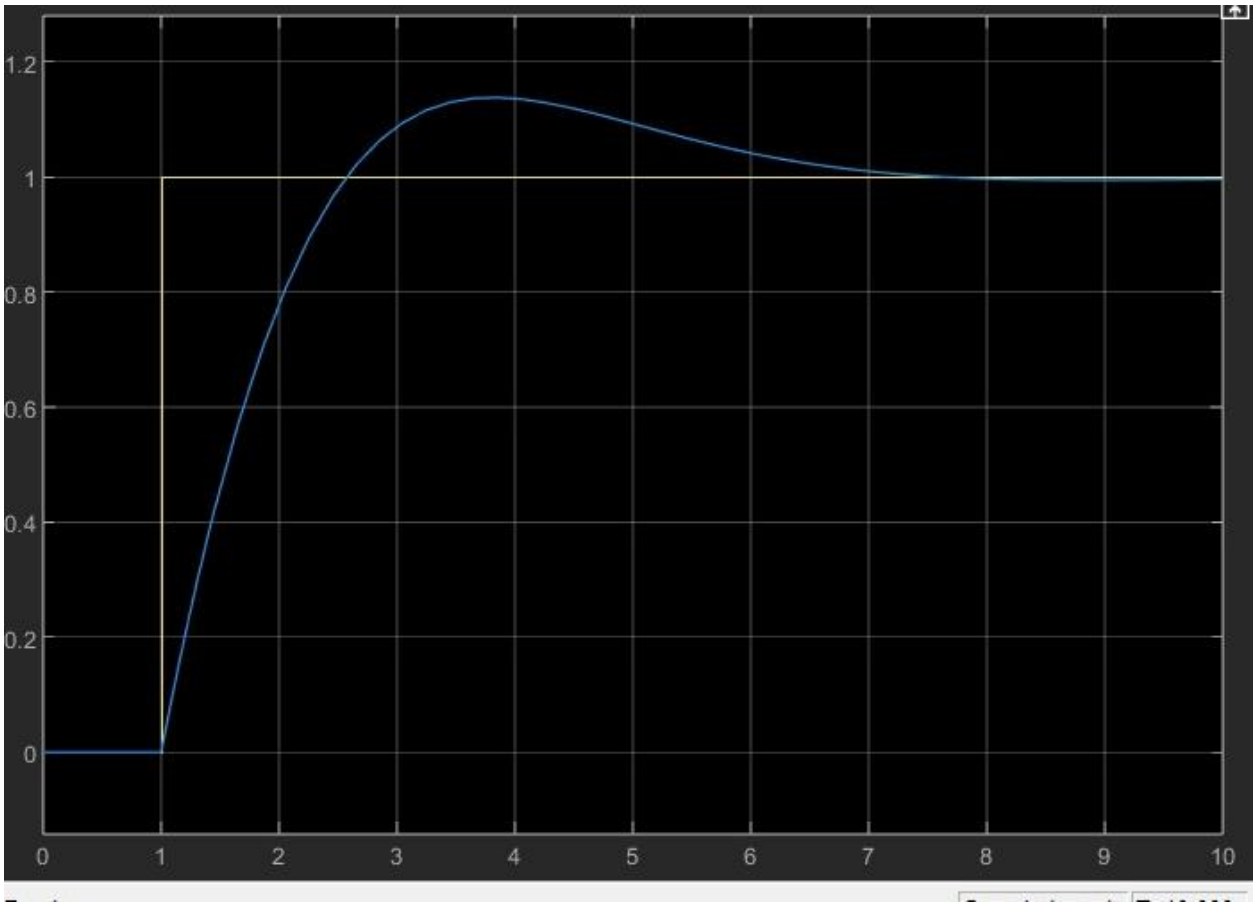
Graph of Velocity



OBSERVATIONS

Part _02

Graphs of Sun position, position



CONCLUSION

This laboratory practical on MATLAB Simulink really showed how modeling and simulation of control systems can be effectively done. Building a second-order system, analyzing it, and implementing a PID controller were indeed great lessons in dynamics regarding stability and response. There were several challenges during the experimentation, such as parameter configuration and PID tuning, but all were sorted out by a step-by-step approach. Overall, the experiment has reinforced the importance of simulation tools in engineering applications and provided a hands-on understanding of control system design.

DISCUSSION

This experiment aimed to use MATLAB Simulink for designing and simulating control systems. Control systems are essential in engineering applications for managing the behavior of dynamic systems. MATLAB Simulink provides a graphical environment that simplifies modeling and testing of these systems. The experiment involved constructing a second-order system model to simulate the motion of a solar panel and analyzing its motor's electrical dynamics. A PID controller was also implemented to improve the system's performance, demonstrating the practical application of control theory in solar tracking systems.

During the experiment, various challenges arose. One main challenge involved the setup parameters of the system, the damping constant, and the moment of inertia, knowing how each affected the model took some effort. Assigning value attempts mostly led to incorrect model output; thus, repeated adjustment was required. The use of subsystems to group the model was also not easy, since wrong block connections often led to the failure of the simulation.The most challenging thing was tuning the PID controller since the proportional and integral values had to be tuned such that the system could get stabilized and the desired response could be achieved. Many times, small changes in those values resulted in noticeable variations in the system's behavior. Also, simulation outputs needed accuracy, and errors needed troubleshooting, which required time and concentration.

These challenges were overcome by systematically solving the problems and following the instructions provided. In setting up the parameters, the theoretical equations in the experiment were used for reference to give realistic values. These values were iteratively tested to optimize the performance of the model. The subsystems were created by using the "Create Subsystem" feature in Simulink, which helped to group similar components and reduce complexity. Fine-tuning of the PID controller was done by starting with simple proportional and integral gains and adjusting them in steps while observing the scope outputs. The derivative term was not necessary since the movement of the solar panel is slow and smooth; hence, there is no need to worry about oscillations. These steps ensured that the controller provided a smooth and stable response.

MATLAB Simulink proved to be a powerful tool for this experiment. The graphical interface made it much easier to model and visualize complex systems. With scopes and other features included, analysis of the system behavior in real time was allowed, which really allowed us to study transient responses and system stability more easily. Its flexibility allowed us to change parameters and try various system configurations without heavy coding. These features made it highly effective for learning and testing control systems.

Meanwhile, a number of disadvantages were present in Simulink too. Its complexity is somewhat fearsome for the new users; mastering how to establish models and hunt for mistakes takes some time. Quite often, mistakes in block configurations or parameter values developed into long debugging. Furthermore, simulations with large models or detailed dynamics could stress out system resources, resulting in a slower performance on less powerful computers.

The behavior of the solar panel motion was observed using the position and velocity graphs, as shown in the observations. The position graph demonstrated a gradual increase, reflecting the steady rotation of the solar panel in response to the applied torque. The velocity graph showed an initial spike followed by stabilization, indicating that the system quickly reached a steady state after a brief transient period. These graphs highlight the effectiveness of the control system in managing the motion of the solar panel. The smooth stabilization in both position and velocity confirms that the system is well-damped, ensuring reliable performance without oscillations or instability.

The following experiment has successfully demonstrated the capability of MATLAB Simulink in modeling and analyzing control systems. We can get a second-order system model and then add a PID controller to obtain practical experience in designing and optimizing dynamic systems. Parameter setup and controller tuning in the experiment had several problems that were solved through systematic approaches and experimentation. Simulink's visual interface and flexibility made it an indispensable tool for understanding control system behavior, despite its learning curve. Overall, this practical session has reinforced theoretical concepts and provided valuable insights into real-world applications like solar tracking.

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